Pathology of the Labia, Cervix, and Vagina

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Block: Endocrine/Urology/Reproduction
Thread: Pathology

Conflict of Interest: Dr. Chan has no conflicts of interest to disclose.

Textbooks

Robbins and Cotran: Pathologic Basis of Disease, 8th ed.—Chapter 22: The Female Genital Tract

Rubin's Pathology: Mechanisms of Human Disease, 8th ed.—Chapter 24: The Female Reproductive System and Peritoneum

Learning Objectives

By exam time, you should be able to do the following:

1. Describe the role of HPV in the pathogenesis of gynecologic tumors.
2. Describe and identify the histopathologic features of HPV-related cervical dysplasia.
3. Explain and describe the relationship between inflammatory processes of the vulva and malignancy.
4. Describe and identify infectious/inflammatory processes of the vulva and cervix.

Study Questions

1. What are the histologic characteristics of HPV infection in the cervix? What are the morphological differences between a low-grade and high-grade squamous intraepithelial lesion (SIL)? What is the relationship between a SIL and developing invasive squamous cell carcinoma?

2. What is the histologic hallmark of lichenoid inflammation in the vulva? Is it specific? What malignancies are associated with lichen sclerosis?

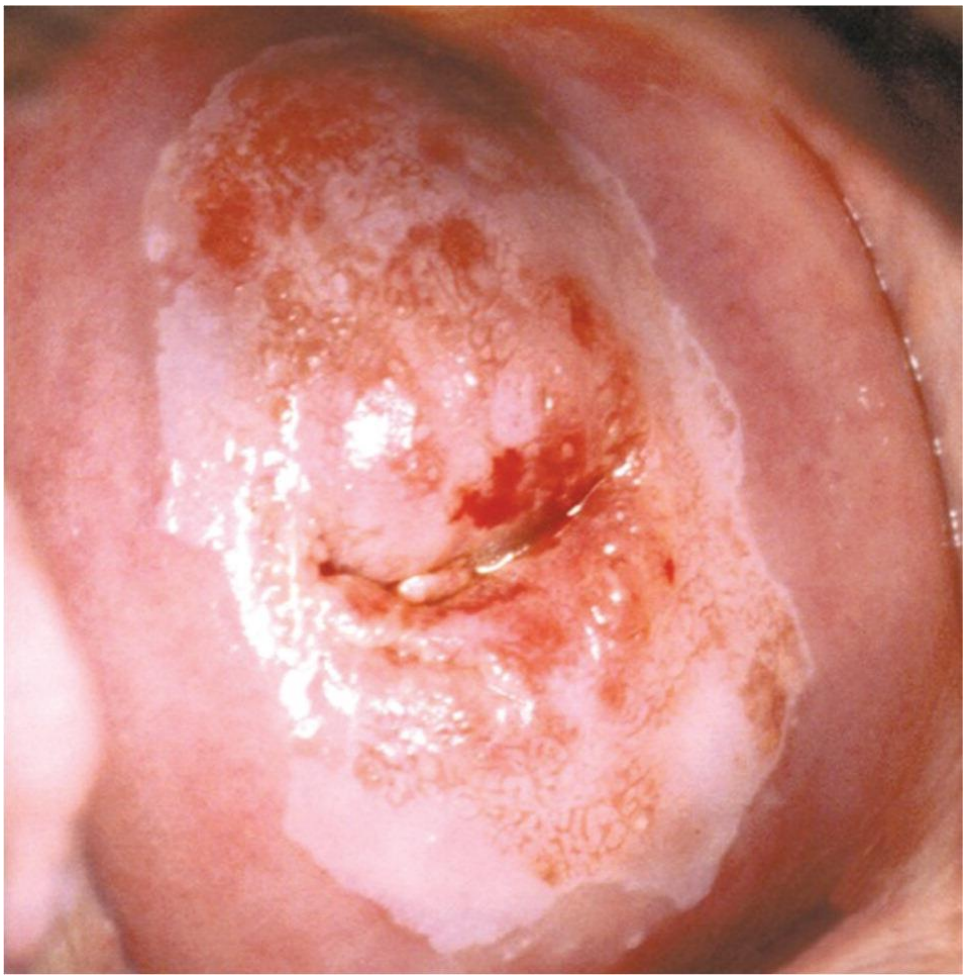
3. What are the histologic findings in fungal infection? What clinical scenarios predispose a patient to fungal infections?

4. What are the classic findings in herpes infections?

Note: You may also refer to Dr. Macnow's Histology of the Ovum and Organs of the Ovum Pathway lab session for a more extensive review of the histology of the reproductive tract.

A. CASE 1

A 34-year-old patient presents to their gynecologist for a colposcopic exam after having an "ASCUS" Pap test result. Their HPV cotesting is positive for high-risk HPV.



1. Based on their HPV cotesting and colposcopy, what is your differential diagnosis?
2. Review the biopsy taken from the abnormal areas of the cervix (S16-28738 1A, LINK) and compare to normal cervix (S18-19864 A1, LINK), taking special note of:
    a. Nucleus:cytoplasm (N:C) ratio
    b. Basal maturation defect
    c. Nuclear atypia

   d. Disruptions in the basement membrane
   e. Relationship between the atypical epithelium and the underlying stroma
3. What is your differential diagnosis? What is the natural course of this lesion if left untreated? How is this lesion related to their HPV status?

The patient is lost to follow-up for 4 years. They present again with an "atypical" Pap test and undergo a second colposcopy with associated biopsies.

4. Review S16-20743 1A (LINK). Again, take particular note of:
   a. Nucleus:cytoplasm (N:C) ratio
   b. Basal maturation defect
   c. Nuclear atypia
   d. Disruptions in the basement membrane
   e. Relationship between the atypical epithelium and the underlying stroma
5. What is your differential diagnosis? Now what is the natural course of this lesion if left untreated?

The patient is lost to follow-up for a second time. They return ten years later with vaginal bleeding. On clinical exam, they have an ulcerated cervical mass with extension into the parametrial soft tissue. The clinical decision is made to have a cold knife cone excisional procedure, which shows high-grade squamous intraepithelial lesion with associated invasive squamous cell carcinoma.

6. Review S15-13905 1A (LINK). Notice the:
   f. Nucleus:cytoplasm (N:C) ratio
   g. Basal maturation defect
   h. Nuclear atypia
   i. Disruptions in the basement membrane
   j. Relationship between the atypical epithelium and the underlying stroma
7. Now what is your differential diagnosis?

B. CASE 2

A 27-year-old patient presents to their gynecologist with a chief complaint of a two-week history of vaginal itching. They have had two weeks of vaginal itching with thick white clumpy discharge. They have had no other complaints, although they do note having had an upper respiratory infection treated with antibiotics a month ago. Prior to this episode, their gynecologic visits were unremarkable. Their gynecologist does a wet prep (see cervical image on next slide), checks the pH, and happens to take a cervical biopsy.

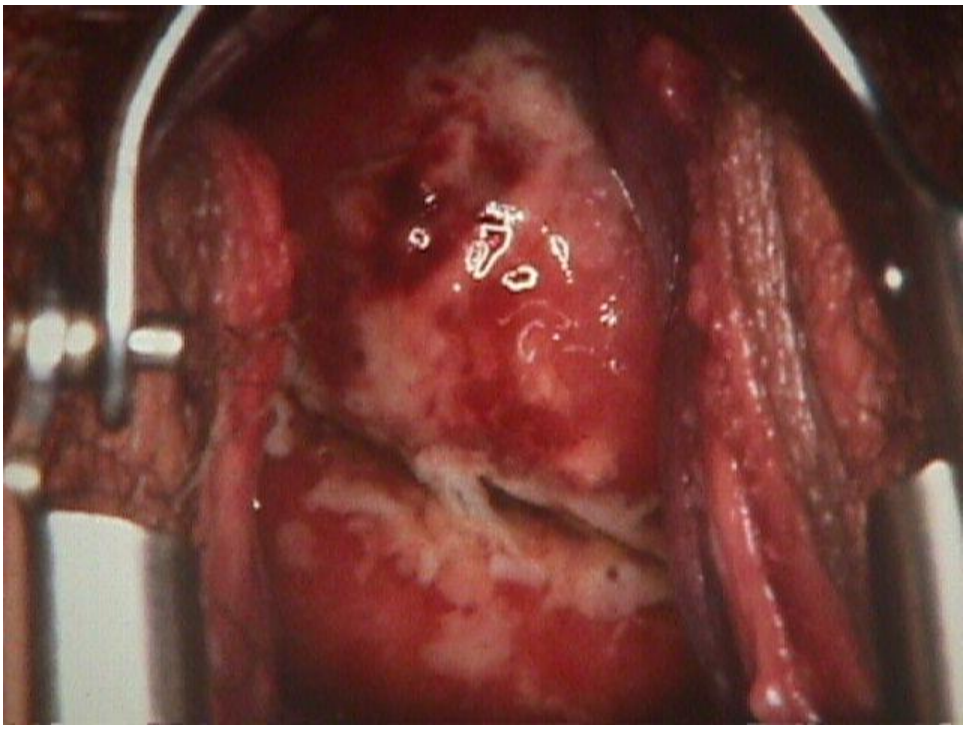


1. Based on their clinical history, what is your differential diagnosis?
2. Examine the patient's cervical biopsy (S17-2518 1A, LINK), looking specifically at:
   a. Presence of lack thereof of dysplasia
   b. Presence of acute inflammatory cells (neutrophils)
   c. Presence of fungal elements
3. Special stains for silver highlight fungal organisms. Confirm your diagnosis with Gomori methenamine silver (GMS) stain (S17-2518 1A GMS, LINK).
4. What is your diagnosis? What organism is the GMS most likely highlighting? What are some risk factors for this disease process?

C. CASE 3

A 32-year-old patient presents to their gynecologist with a "painful rash" of the vulva for the last three days. They describe a three-day history of a diffuse burning, itching sensation. They also complain of a headache and general sense of feeling "unwell."

A colposcopic view of the patient's cervix:



1. Review the patient's cervical biopsy (S16-5420, LINK), taking special note of:
   a. Presence of inflammatory cells
   b. Presence of nuclear atypia
   c. Presence of "normal" epithelium
2. Immunohistochemical stains can highlight specific microorganisms. In this case, an immunohistochemical stain for HSV-1 and -2 was done (S16-5420 HSV1+2, LINK).
3. What is your diagnosis? What are some risk factors for this disease process?

D. CASE 4

A 66-year-old patient presents to their gynecologist with a chief complaint of vulvar itching. They are not sure how long they have had the itching, but they think it got worse after menopause. On physical exam (see image below), their vulva is thin and fragile appearing, and the gynecologist takes a biopsy.



1. Based on the clinical presentation, what is your differential diagnosis?
2. Compare a normal vulva (S15-31652 1M, LINK) to the patient's biopsy (S16-26259, LINK), specifically noting:
    a. Thickness of epithelial layer
    b. Presence or lack thereof of rete ridges
    c. Homogenization of the dermal layer
3. What is your diagnosis? What is this disease process a risk factor for?

On follow-up a year later, their vulva is still thin and fragile; however, there are nodular areas, which the gynecologist biopsies.

4. Examine S15-31652 (LINK). Compare the current biopsy with the prior biopsy, taking particular note of:
    a. Similarity to prior biopsy specimen
    b. Area of thickened epithelium with nuclear atypia
    c. Relationship between atypical epithelium and underlying stroma
5. Now what is your diagnosis? How is this different from the other squamous cell carcinoma case?

Answers to Study Questions

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

1. What are the histologic characteristics of HPV infection in the cervix? What are the morphological differences between a low-grade and high-grade squamous intraepithelial lesion (SIL)? What is the relationship between a SIL and developing invasive squamous cell carcinoma?

   *HPV is characterized by **koilocytic atypia** with perinuclear halos, hyperchromasia, and nuclear pleomorphism. Low-grade and high-grade lesions are differentiated by the presence and extent of **squamous maturation defect**. Low-grade lesions often have abundant koilocytic change, but minimal maturation defect, whereas high-grade lesions often show a decreased nuclear:cytoplasmic ratio with a marked maturation defect. Low-grade squamous intraepithelial lesions (LSIL) do not progress directly to invasive carcinoma, and generally regress on their own, with 10% progressing to high-grade squamous intraepithelial lesions (HSIL). High-grade squamous intraepithelial lesions are considered premalignant, with the majority persisting, and 10% progressing to carcinoma within the next 2–10 years.*

2. What is the histologic hallmark of lichenoid inflammation in the vulva? Is it specific? What malignancies are associated with lichen sclerosis?

   *Lichenoid inflammation is defined by a **dense band-like lymphocytic infiltrate in the dermis**. It is associated with multiple inflammatory conditions, including lichen simplex and lichen sclerosis. Although lichen sclerosis is not considered premalignant in of itself, patients with lichen sclerosis have a slightly increased chance of developing squamous cell carcinoma of the vulva.*

3. What are the histologic findings in fungal infection? What clinical scenarios predispose a patient to fungal infections?

   *Fungal infections are often marked by an **acute inflammatory response in the epithelium**. Although yeast is part of normal vaginal flora, symptomatic candidiasis generally occurs as a response to a disturbance in the vaginal microbial system. These disturbances include prolonged antibiotic use and douching.*

4. What are the classic findings in herpes infections?

Herpetic lesions are generally biopsied in the ulcerative phase. The epithelium is generally denuded with ***marked acute inflammation*** *in the ulcer bed. The inflammatory exudate may show changes characteristic of herpes simplex virus including* ***multinucleated squamous cells with viral inclusions known as Cowdry bodies****.*